

QUESTION BANK

Course Code: 21EIC552J **Course Title:** Industrial Process Automation

MODULE 1: Automation System Structure & Control Strategies

1. List the evolution of automation technology over the past decades.
2. Define "Physical process" and name its categories.
3. What is a process signal? List the types of physical signals.
4. List the benefits of Automation.
5. What is called an automation function?
6. Draw the basic structure of an automation system.
7. Draw the typical automation cycle.
8. What is a control subsystem and what are its functions?
9. What are the tasks performed by the Control Subsystem?
10. Discuss in detail about Input, Output, and Human Interface Instrumentation Subsystems with an example.
11. Draw the structure of an analog instrumentation device and explain its measurement and control mechanism.
12. Write the basic function of the human interface system.
13. Illustrate the interfacing of the control subsystem with the instrumentation subsystem with an example.
14. Describe the Process Management system through the schematic and operation of the unattended/attended/automated water heater system.
15. Explain briefly about the automation steps in the water-heating process.
16. Define delay time in a process control.
17. What is meant by steady-state error?
18. List the basic strategies for implementing automation schemes.
19. Discuss various types of closed-loop control schemes for continuous process automation.
20. Illustrate with a neat sketch on Feed-Forward and Cascade control schemes.

MODULE 3: SCADA, DCS, and Distributed Control

1. Explain briefly about various functional elements of a SCADA system.
2. Discuss briefly about the functional elements of a Remote Terminal Unit (RTU).
3. List the signals that come into the Remote Terminal Unit (RTU).
4. List the signals that leave the Remote Terminal Unit (RTU).
5. What are the main functions of a Master Terminal Unit (MTU) of a SCADA?
6. Describe how the MTU handles discrete control instructions to open and close a valve in a SCADA system.
7. Describe how the MTU handles analog control instructions to open and close a valve in a SCADA system.

8. Explain briefly about the analog input and output module of a SCADA system.
9. Explain briefly about the digital input and output module of a SCADA system.
10. Many protocols divide the binary message into three fields. What are they?
11. Describe briefly about the architecture of a DCS (Distributed Control System).
12. Define the control complexity ratio of a control system structure.
13. Compare different control system architectures based on their features.
14. Explain the architectural parameters of LCU (Local Control Unit) and compare its different configurations.
15. Sketch the architecture of a geographically centralized control configuration.
16. Describe briefly about the hybrid system architecture.
17. Explain the design considerations and requirements of the operator interface in detail.
18. Recall the use of function blocks in programming LCU.
19. List a few signal status functions used in a function block.
20. Explain the function block programming approach with an example.